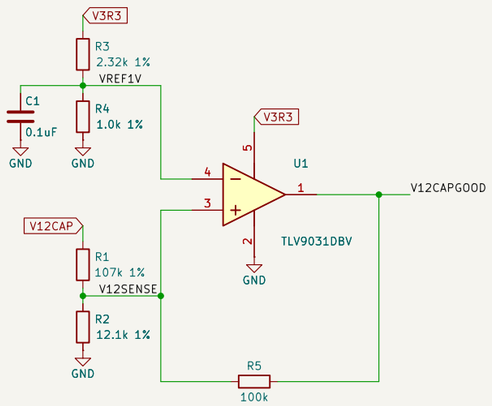
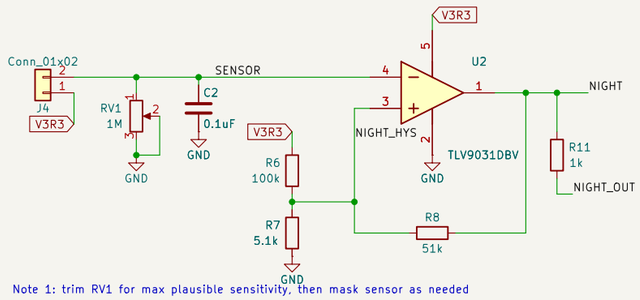
V12CAP Monitor (U1)

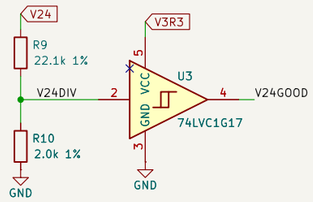


Night Sensor (U2, J4)

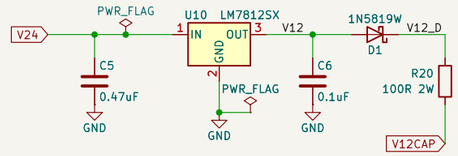


Note 1: trim RV1 for max plausible sensitivity, then mask sensor as needed

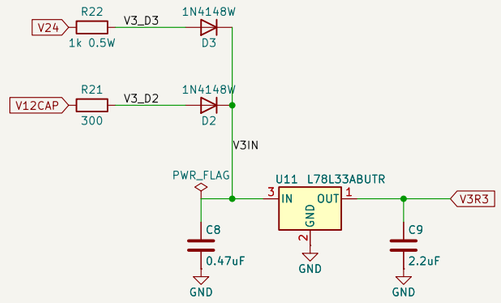
V24 Detect (U3)



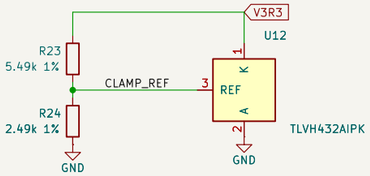
12V Regulator (U10)



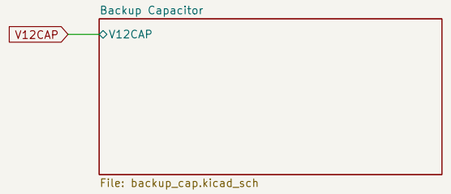
3V3 Regulator (U11)



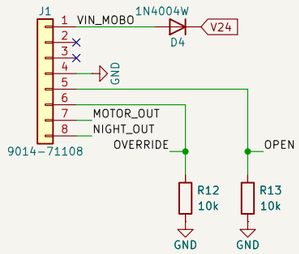
3V3 Clamp (U12)



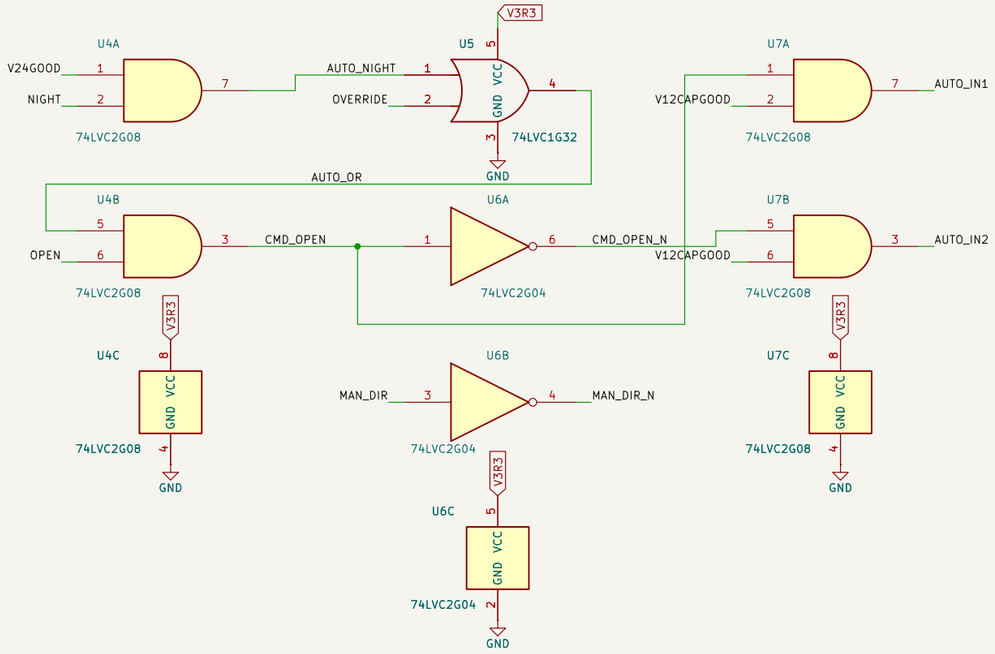
Backup Capacitor (sheet)



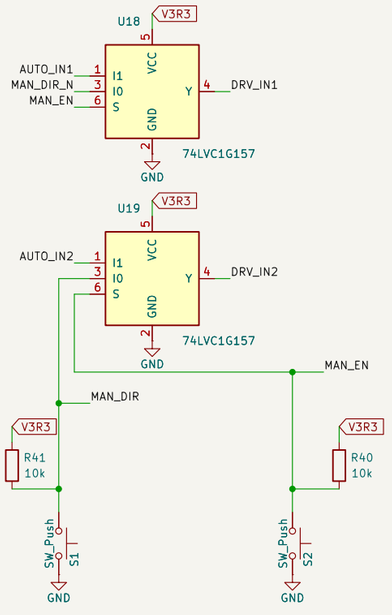
MOBO Connector (J1)



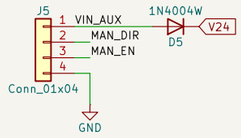
Control Logic (U4–U7)



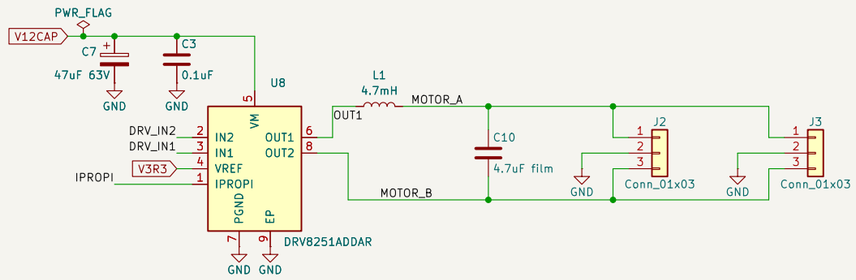
Manual/Auto Select (U18, U19)



Aux Power / Manual (J5)



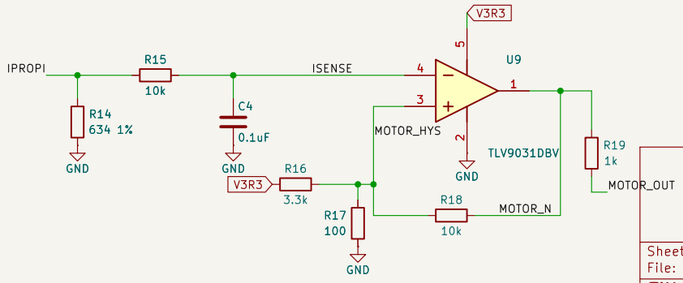
Motor Driver (U8, J2, J3)



DRV8251A truth table

IN1	IN2	OUT1	OUT2
0	0	Hi-Z	Hi-Z
0	1	L	H
1	0	H	L
1	1	L	L

Current Sense (U9)



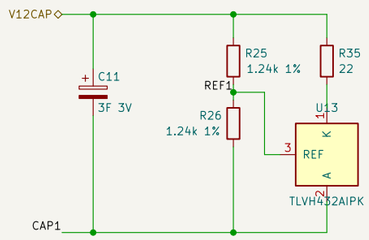
Note 2: IPROPI Iout = 1.58 mA/A; node internally clamped to V<=Vref; motor current limits when V=Vref, here Ilim=3.3A

Sheet: /
File: shuttercontrol.kicad_sch

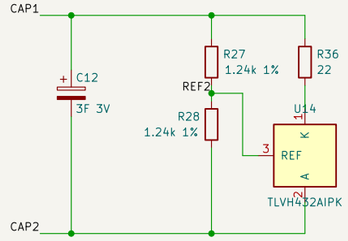
Title: New Shutter Controller

Size: A3	Date: 2026-10-11	Rev: v.02h
KiCad E.D.A. 10.0.7		Id: 1/2

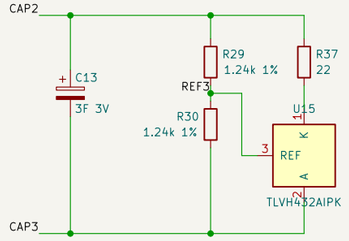
Cell 1 (U13, C11)



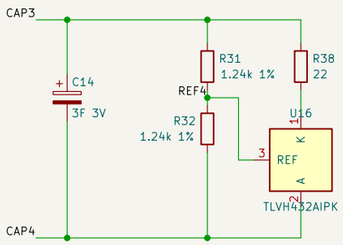
Cell 2 (U14, C12)



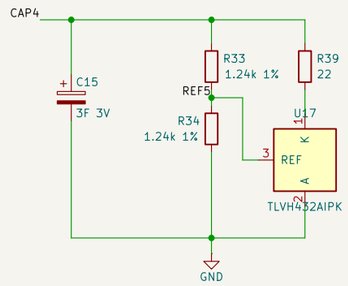
Cell 3 (U15, C13)



Cell 4 (U16, C14)



Cell 5 (U17, C15)



Sheet: /Backup Capacitor/		
File: backup_cap.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 2/2